Chapter 8 — Chemical Equilibrium

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 181–184

Q.1 — Multiple Choice Questions (with correct option and reason)

I. For which system does Kc have units of (concentration)$^{-1}$?

(a) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ **(b)** $H_2 + I_2 \rightleftharpoons 2HI$

(c) $2NO_2 \rightleftharpoons N_2O_4$ **(d)** $2HF \rightleftharpoons H_2 + F_2$

Answer: (c) $2NO_2 \rightleftharpoons N_2O_4$ — For $2NO_2 \rightleftharpoons N_2O_4$, $K_c = [N_2O_4]/[NO_2]^2$, giving units $(\text{mol dm}^{-3})^1$.

II. For $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -197.8$ kJ mol^{-1}, which statement is correct?

(a) The value of Kp falls with a rise in temperature. **(b)** The value of Kp falls with increasing pressure.

(c) Adding V_2O_5 catalyst increases the equilibrium yield of SO_3. **(d)** The value of Kp is equal to Kc.

Answer: (a) The value of Kp falls with a rise in temperature. — The forward reaction is exothermic, so raising temperature shifts equilibrium left and decreases Kp.

III. For $2SO_3(g)$ $2SO_2(g) + O_2(g)$, the correct Kc expression is:

(a) $[SO_2]^2/[SO_3]$ **(b)** $[SO_2]^2[O_2]/[SO_3]^2$

(c) $[SO_3]^2/[SO_3]^2[O_2]$ **(d)** $[SO_2][O_2]$

Answer: (b) $[SO_2]^2[O_2]/[SO_3]^2$ — $K_c = [SO_2]^2[O_2]/[SO_3]^2$.

IV. A saturated solution is a dynamic equilibrium; macroscopically the dissolved concentration is constant because:

(a) No more solute particles are dissolving. **(b)** The rate of dissolution is zero.

(c) Solute particles are dissolving and precipitating at the same rate. **(d)** All solute particles have dissolved.

Answer: (c) Solute particles are dissolving and precipitating at the same rate. — At equilibrium the rates of dissolution and crystallisation are equal.

V. Which statement correctly describes the effect of temperature on the equilibrium constant?

(a) Kc is directly proportional to temperature. **(b)** Kc is inversely proportional to temperature.

(c) Kc depends on the enthalpy change of the reaction. **(d)** Temperature has no effect on Kc.

Answer: (c) Kc depends on the enthalpy change of the reaction. — The direction of the change in K with temperature depends on whether the reaction is endothermic or exothermic (ΔH).

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VI. For $2H_2O(g) \rightleftharpoons 2H_2(g) + O_2(g)$ (forward endothermic), which change decreases the equilibrium amount of H_2O?

(a) Adding more oxygen
(b) Adding a solid phase catalyst
(c) Decreasing the volume of the container
(d) Increasing the temperature at constant pressure

Answer: (d) Increasing the temperature at constant pressure — Raising temperature favours the endothermic forward reaction, consuming H_2O.

VII. Kc = 0.040 at 450 °C for $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. Evaluate Kp.

(a) 0.40
(b) 0.64
(c) 2.4
(d) 0.052

Answer: (c) 2.4 — Kp = Kc(RT)^Δn; Δn = 1, T = 723 K, R = 0.0821 → Kp = 0.040 × 0.0821 × 723 ≈ 2.4.

VIII. In which gaseous equilibrium does pressure have no effect on the equilibrium position?

(a) $2NO_2(g) \rightleftharpoons N_2O_4(g)$
(b) $PCl_5(g)$ $PCl_3(g) + Cl_2(g)$
(c) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$
(d) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

Answer: (c) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ — Δn(gas) = 0 for $CO + H_2O \rightleftharpoons CO_2 + H_2$, so pressure does not shift the equilibrium.

IX. For $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$, if $[H_2O]$ is increased, $[H_2]$ and $[O_2]$ will:

(a) Increase
(b) Decrease
(c) Remain the same
(d) Change irregularly

Answer: (a) Increase — Adding product shifts equilibrium to the left, increasing the reactants.

X. For a specific reaction, the value of Kc:

(a) Always remains the same at different reaction conditions.
(b) Increases if the concentration of one product is increased.
(c) Changes with changes in the temperature.
(d) Increases if the concentration of one reactant is increased.

Answer: (c) Changes with changes in the temperature. — Kc depends only on temperature for a given reaction.

Q.2 — Short Answer Questions

a. What is meant by the state of chemical equilibrium?

Chemical equilibrium is the state of a reversible reaction in which the rates of the forward and reverse reactions are equal, so the concentrations of reactants and products remain constant with time.

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b. Define reversible reaction. Give an example.

A reversible reaction proceeds in both forward and reverse directions under the same conditions. Example: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.

c. The change of volume disturbs the equilibrium position for some gas-phase reactions but not the equilibrium constant.

Changing volume changes concentrations/partial pressures and can shift the equilibrium position toward the side with fewer or more gas moles (Le Chatelier). However, Kc/Kp depends only on temperature, so at constant temperature the equilibrium constant remains unchanged.

d. Mention the characteristics of chemical equilibrium.

It is dynamic (both reactions continue); rates of forward and reverse reactions are equal; concentrations remain constant; it can be approached from either direction; it is attained only in a closed system; and a catalyst speeds attainment but does not change the equilibrium position.

e. Reversible reaction attains equilibrium which is dynamic, not static. Explain.

At equilibrium the forward and reverse reactions continue at equal rates, so there is no net change in concentrations. Because molecular-level reactions do not stop, the equilibrium is dynamic rather than static.

f. Why do the rates of forward reactions slow down when a reversible reaction approaches equilibrium?

As reactants are consumed their concentrations decrease, so the forward rate falls. Meanwhile product concentrations rise and the reverse rate increases until the two rates become equal.

g. Why can ice at 0 °C be melted by applying pressure without supply of heat from outside?

Ice is less dense than water, so applying pressure favours the denser liquid phase (Le Chatelier). The melting point of ice decreases under pressure, allowing it to melt at 0 °C without external heating.

h. Write two conditions of equilibrium constant.

(i) The value of K is constant at a fixed temperature for a given reaction. (ii) K is written for the balanced equation as written; pure solids and pure liquids are omitted from the expression.

i. For $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, initial $[SO_2] = 0.060$ and $[O_2] = 0.050$ mol dm^{-3}; at equilibrium $[SO_3] = 0.040$ mol dm^{-3}. Find equilibrium $[O_2]$.

From stoichiometry, forming 0.040 mol dm^{-3} SO consumes 0.020 mol dm^{-3} O_2 (2:1 ratio). Equilibrium $[O_2] = 0.050 - 0.020 = 0.030$ mol dm^{-3}.

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Descriptive / Long Questions

Q.3 (a) Define and explain the law of mass action and derive the expression for the equilibrium constant. (b) Write Kc expressions for the given reactions.

(a) The law of mass action states that the rate of a reaction is proportional to the product of the active masses (molar concentrations) of the reactants, each raised to its stoichiometric coefficient. For aA + bB ⇌ cC + dD, at equilibrium the forward and reverse rates are equal, giving Kc = [C]^c[D]^d / [A]^a[B]^b.

(b) (i) $Sn^{2+} + 2Fe^{3+} \rightleftharpoons Sn^{4+} + 2Fe^{2+}$: $K_c = [Sn^{4+}][Fe^{2+}]^2 / ([Sn^{2+}][Fe^{3+}]^2)$.

(ii) $Ag^{+} + Fe^{2+} \rightleftharpoons Fe^{3+} + Ag(s)$: $K_c = [Fe^{3+}] / ([Ag^{+}][Fe^{2+}])$ (Ag solid omitted).

(iii) $N_2 + O_2 \rightleftharpoons 2NO$: $K_c = [NO]^2 / ([N_2][O_2])$.

(iv) $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$: $K_c = [NO]^4[H_2O]^6 / ([NH_3]^4[O_2]^5)$.

(v) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$: $K_c = [PCl_3][Cl_2] / [PCl_5]$.

Q.4 Write Kc for the following reactions, with reaction mixture volume V dm³.

(i) $CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$: $K_c = [ester][H_2O] / ([acid][alcohol])$. Since Δn = 0, V cancels.

(ii) $2HI \rightleftharpoons H_2 + I_2$: $K_c = [H_2][I_2] / [HI]^2$; V cancels.

(iii) $N_2 + 3H_2 \rightleftharpoons 2NH_3$: $K_c = [NH_3]^2 / ([N_2][H_2]^3)$. In terms of moles n and volume V: $K_c = (n_NH_3)^2 V^2 / (n_N_2 \cdot n_H_2^3)$.

Q.5 For $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $\Delta H = +90\ kJ\ mol^{-1}$, what is the effect of: (i) increasing temperature (ii) decreasing volume (iii) adding catalyst (iv) adding chlorine?

(i) Increasing temperature favours the endothermic forward reaction → more PCl_3 and Cl_2 (equilibrium shifts right).

(ii) Decreasing volume (increasing pressure) favours the side with fewer gas moles (left, 1 mol) → shifts left.

(iii) A catalyst speeds up both forward and reverse reactions equally → no change in equilibrium position.

(iv) Adding Cl_2 (a product) shifts equilibrium left → more PCl_5 forms.

Q.6 Synthesis of ammonia by Haber's process is exothermic: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = -92.46\ kJ$. What is the effect of change of temperature at equilibrium?

Because the forward reaction is exothermic, increasing temperature shifts equilibrium to the left (toward N_2 and H_2), decreasing the yield of NH_3 and decreasing K. Lowering temperature shifts equilibrium right and increases NH_3 yield, but the rate becomes slow; hence a compromise temperature (~450 °C) is used industrially.

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Numerical Problems (fully worked)

Q.7 Kc = 0.016 at 520 °C for $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$. Equilibrium mixture: [HI] = 0.08, [H_2] = 0.01, [I_2] = 0.01 M. More HI is added so [HI] becomes 0.096 M. Find the new equilibrium concentrations.

Given: Kc = 0.016; after addition $[HI]_0 = 0.096$, $[H_2]_0 = [I_2]_0 = 0.010$ M

Formula: $K_c = [H_2][I_2]/[HI]^2$; let x = amount of HI that dissociates

At new equilibrium: $[HI] = 0.096 - 2x$, $[H_2] = [I_2] = 0.010 + x$.

$0.016 = (0.010 + x)^2/(0.096 - 2x)^2$. Taking square root: $0.1265 = (0.010 + x)/(0.096 - 2x)$.

$0.010 + x = 0.01214 - 0.253x \rightarrow 1.253x = 0.00214 \rightarrow x = 0.00171$ M.

$[HI] = 0.096 - 2(0.00171) = 0.0926$ M; $[H_2] = [I_2] = 0.010 + 0.00171 = 0.0117$ M.

Answer: [HI] ≈ 0.0926 M; [H_2] = [I_2] ≈ 0.0117 M

The book lists this under Descriptive Questions; it is solved here as a numerical.

Q.8 Kc for $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$ is 4. A mixture of 3 mol acid and 1 mol alcohol reaches equilibrium. Calculate the amount of ethyl acetate at equilibrium.

Given: Initial acid = 3 mol, alcohol = 1 mol, ester = water = 0; Kc = 4

Formula: $K_c = x^2 / [(3 - x)(1 - x)]$ (volume cancels)

Let x mol of ester form. At equilibrium: acid = 3 − x, alcohol = 1 − x, ester = water = x.

$4 = x^2/[(3 - x)(1 - x)] \rightarrow 4(3 - 4x + x^2) = x^2 \rightarrow 3x^2 - 16x + 12 = 0$.

$x = [16 \pm \sqrt{(256 - 144)}]/6 = [16 \pm 10.58]/6$. The valid root ($x < 1$) is x = 0.903 mol.

Answer: Ethyl acetate at equilibrium ≈ 0.90 mol

Q.9 For $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$: (i) write Kp. (ii) 1.00 mol steam and 1.00 mol CO reach equilibrium; 33.3% of the mixture is hydrogen. Calculate Kp and state its units.

Given: Initial H_2O = CO = 1.00 mol; at equilibrium H_2 = 33.3% of total moles

Formula: $K_p = (p_{H_2} \cdot p_{CO_2})/(p_{H_2O} \cdot p_{CO})$

Total moles remain 2.00 ($\Delta n = 0$). Moles $H_2 = 0.333 \times 2.00 = 0.666$ mol = CO_2.

Moles H_2O = CO = 1.00 − 0.666 = 0.334 mol.

Since total pressure and moles cancel, $K_p = (0.666 \times 0.666)/(0.334 \times 0.334) \approx 3.98 \approx 4.0$.

$\Delta n(gas) = 0$, so Kp has no units.

Answer: Kp ≈ 4.0 (dimensionless)

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Quick Check Solutions (in-chapter practice boxes)

Quick Check 8.1 (book p. 166)

a. Differentiate between macroscopic and microscopic events.

Macroscopic events are bulk observations (colour, pressure, concentration) that appear constant at equilibrium. Microscopic events are the molecular-level forward and reverse reactions, which continue even at equilibrium.

b. Why equilibrium is dynamic in terms of microscopic events.

At equilibrium the forward and reverse reactions continue at equal rates, so there is no net macroscopic change, but molecules are still reacting.

c. For $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$, products are higher at equilibrium. Plot concentration vs time and rate vs time.

Concentration–time graph: reactant concentrations fall and product concentrations rise, then all become constant (products at a higher level). Rate–time graph: forward rate falls and reverse rate rises until they become equal and constant.

Quick Check 8.2 (book p. 167)

a. Justify that dynamic equilibrium exists between water and its vapour at 100 °C.

At 100 °C (at 1 atm) the rate of evaporation equals the rate of condensation, so the amounts of liquid and vapour remain constant while both processes continue.

b. Does dynamic equilibrium exist between ice and water at 0 °C? Explain.

Yes. At 0 °C the rate of melting equals the rate of freezing, so the amounts of ice and water remain constant while both processes continue.

Quick Check 8.3 (book p. 169)

a. Differentiate homogeneous and heterogeneous equilibria.

Homogeneous equilibrium: all reactants and products are in the same phase. Heterogeneous equilibrium: reactants and products are in different phases.

b. Why the forward rate slows as a reversible reaction approaches equilibrium.

Reactant concentrations decrease as they are consumed, so the forward rate falls until it equals the reverse rate.

c. Soda water equilibrium $CO_2(g) + H_2O(l) \rightleftharpoons HCO^-(aq) + H^+(aq)$: (i) direction when bottle opened (ii) why closed cap is needed (iii) direction when cap removed.

(i) Opening lowers the CO_2 pressure, so equilibrium shifts left (toward CO_2 gas) and gas escapes. (ii) A closed cap maintains the CO_2 pressure needed for equilibrium. (iii) Removing the cap shifts the equilibrium left, releasing CO_2 (fizzing).

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Quick Check 8.4 (book p. 172)

a. Write Kc for: (i) $Sn^{2+} + 2Fe^{3+} \rightleftharpoons Sn^{4+} + 2Fe^{2+}$ (ii) $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$ (iii) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$.

(i) $K_c = [Sn^{4+}][Fe^{2+}]^2/([Sn^{2+}][Fe^{3+}]^2)$. (ii) $K_c = [NO]^4[H_2O]^6/([NH_3]^4[O_2]^5)$. (iii) $K_c = [PCl_3][Cl_2]/[PCl_5]$.

b. Calculate Kc for $H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$. Initial $[H_2] = [CO_2] = 10.00$ M; equilibrium $[CO] = 9.47$ M.

At equilibrium $[H_2O] = [CO] = 9.47$ M; $[H_2] = [CO_2] = 10.00 - 9.47 = 0.53$ M. $K_c = (9.47 \times 9.47)/(0.53 \times 0.53) = 319$.

Quick Check 8.5 (book p. 175)

a. Compare Kc and Kp for (i) ammonia synthesis (ii) dissociation of PCl_5.

(i) $N_2 + 3H_2 \rightleftharpoons 2NH_3$: $\Delta n = -2$, so $K_p = K_c(RT)^{-2}$ and $K_p < K_c$. (ii) $PCl_5 \rightleftharpoons PCl + Cl_2$: $\Delta n = +1$, so $K_p = K_c(RT)$ and $K_p > K_c$.

b. $N_2 + 3H_2 \rightleftharpoons 2NH_3$. Total P = 2.0×10^7 Pa; $pN_2 = 1.5\times10^7$ Pa; $pH_2 = 0.4\times10^7$ Pa. Calculate Kp.

$pNH_3 = 2.0\times10^7 - 1.5\times10^7 - 0.4\times10^7 = 0.1\times10^7$ Pa. $K_p = (pNH_3)^2/(pN_2 \cdot pH_2^3) = (0.1\times10^7)^2/[(1.5\times10^7)(0.4\times10^7)^3] = 1.0\times10^{-15}$ Pa^{-2}.

Quick Check 8.6 (book p. 177)

How does the direction change for: (i) $2H_2 + O_2 \rightleftharpoons 2H_2O$ (increasing P) (ii) $2CO + O_2 \rightleftharpoons 2CO_2$ (increasing V) (iii) $N_2O_4 \rightleftharpoons 2NO_2$ (increasing P) (iv) $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (increasing V)?

(i) Increasing P shifts right (fewer gas moles). (ii) Increasing V shifts left (more gas moles). (iii) Increasing P shifts left (fewer gas moles). (iv) Increasing V shifts left (more gas moles). In all cases K is unchanged at constant T.

Quick Check 8.7 (book p. 181)

i. How does the proportion of products change as temperature increases for the three systems?

$N_2 + 3H_2 \rightleftharpoons 2NH_3$ (exothermic): product proportion decreases. $N_2O_4 \rightleftharpoons 2NO_2$ (endothermic): product proportion increases. $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (exothermic): product proportion decreases.

ii. Calculate Kc for the reactions at 600 K.

Using $K_p = K_c(RT)^{\Delta n}$ with $R = 0.0821$ atm dm^3 mol^{-1} K^{-1}, $RT = 49.26$. (1) NH_3 synthesis: $K_p = 3.1\times10^{-3}$, $\Delta n = -2 \rightarrow K_c = K_p(RT)^2 = 7.5$ dm^6 mol^{-2}. (2) N_2O_4 dissociation: $K_p = 1.5\times10^1$, $\Delta n = +1 \rightarrow K_c = K_p/(RT) = 0.30$ mol dm^{-3}. (3) SO_3 formation: $K_p = 3.2\times10^3$, $\Delta n = -1 \rightarrow K_c = K_p(RT) = 1.6\times10^5$ dm^3 mol^{-1}.

Quick Check 8.7 table: the second equation is printed as $N_2O_4 \rightleftharpoons 2NH_2$; it should be $N_2O_4 \rightleftharpoons 2NO_2$.

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Textbook checks flagged in this chapter

• pp. 183–184: Q.7–Q.9 are printed under Descriptive Questions but are numerical; they are fully worked here.

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